

Random Vector and its Distribution

1.1 The additivity of some distributions

Example 1.1.1 ► The additivity of Poisson Distribution

Suppose that $X \sim P(\lambda_1)$, $Y \sim P(\lambda_2)$, and X, Y are independent. Then

$$Z = X + Y \sim P(\lambda_1 + \lambda_2)$$

Example 1.1.2 ► The additivity of Binomial Distribution

Suppose that $X \sim b(n, p)$, $Y \sim b(m, p)$ and X, Y are independent. Then

$$Z = X + Y \sim b(n + m, p)$$

Example 1.1.3 ► the additivity of Normal Distribution

Suppose that $X \sim N(\mu_1, \sigma_1^2)$, $Y \sim N(\mu_2, \sigma_2^2)$ and X, Y are independent. Then

$$Z = X + Y \sim N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$$

Example 1.1.4 ► the additivity of Gamma Distribution

Suppose that $X \sim Ga(\alpha_1, \lambda)$, $Y_2 \sim Ga(\alpha_2, \lambda)$, and X, Y are independent. Then

$$Z = X + Y \sim Ga(\alpha_1 + \alpha_2, \lambda)$$